THE FIBRE-DIMENSION FUNCTION OF A FLAT FINITELY PRESENTED MAP OF SPACES IS LOWER SEMI-CONTINUOUS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7203-7211 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 7213-7224, the complete author proof (10 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side; schemes twin uncounted per ruling.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. In-source lemma-base-change-dimension-fibres; More on Morphisms, lemma dimension-fibres-flat.

Source reference. Original lemma 05F8: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7183–7196. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0D4H: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7224–7224. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a flat morphism of finite presentation of algebraic spaces over S. Let $n_{X/Y}$ be the function on Y giving the dimension of fibres of f introduced in Lemma 05F8. Then $n_{X/Y}$ is lower semi-continuous.*

Proof. Let $V \to Y$ be a surjective étale morphism where V is a scheme. If we can show that the composition $n_{X/Y} \circ |g|$ is lower semi-continuous, then the lemma follows as $|g|$ is open. Hence we may assume Y is a scheme. Working locally we may assume V is an affine scheme. Then we can choose an affine scheme U and a surjective étale morphism $U \to X$. Then $n_{X/Y} = n_{U/Y}$. Hence we may assume X and Y are both schemes. In this case the lemma follows from More on Morphisms, Lemma 0D4H. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.